



Open Source Model Rocket: DESIGN AND SIMULATION

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In the previous article in this series, 'Open Source Model Rocket: Design And Simulation (Part 1),' published in June issue, the model rocket was designed using simulation software and its performance was evaluated through virtual flight tests. Building on that work, Part 2 covers the assembly process, circuit connections, flight

computer programming, and testing, culminating in the construction of a physical model rocket based on the earlier design.

Part 2 focuses on the rocket's electronics and stabilisation system. Sensors measure altitude and detect changes in motion, enabling the controller to actively stabilise the rocket by adjusting the aerodynamic

fins during flight. The controller can also be programmed to modify the rocket's flight path according to a predefined trajectory.

To ensure stable flight, proportional-integral-derivative (PID) tuning is performed for self-balancing and fin stabilisation. This enables the rocket to compensate for wind, turbulence and other external disturbances while maintaining its intended flight trajectory.

While full-scale rockets rely on dedicated flight computers, this model uses a simple microcontroller and straightforward programming to implement the same core functions. During testing, the system transmits real-time flight data and fin-control commands over Wi-Fi within its operating range. For longer-range operation, a dedicated telemetry module can be added to transmit flight data to a ground station or another receiving device.

Fig. 1 shows the assembled open source model rocket with a detachable nose cone, while Fig. 2 shows the author's model rocket prototype alongside its OpenRocket simulator design.

To build this system, the required components are listed in Table 1 and

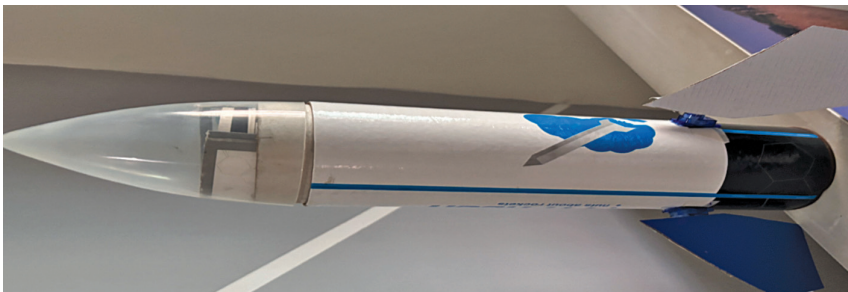


Fig. 1: Assembled open source model rocket with detachable nose cone

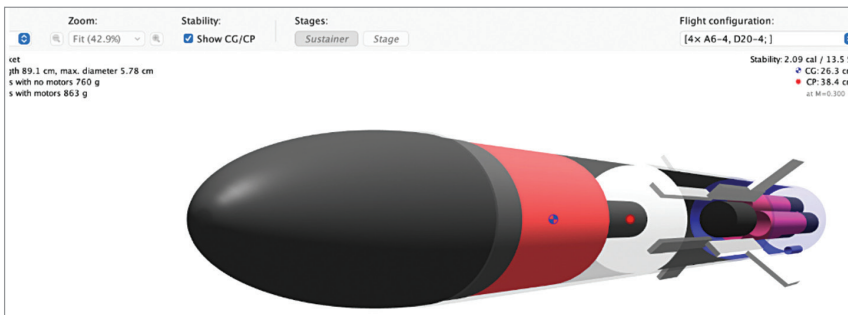


Fig. 2: Author's model rocket prototype and design in the OpenRocket simulator

TABLE 1
BILL OF MATERIALS FOR THE ROCKET BODY

Category	Component	Specification	Recommended Range/Notes
Structure	Body tube	3D-printed tube or a standard commercially available body tube	Diameter and length should match the selected rocket design and corresponding part number.
Structure	Nose cone	3D-printed (PLA/PETG) or a standard commercially available nose cone	Select the dimensions and part number based on the finalised design and simulation results.
Structure	Fins	Three aerodynamic fins	Use three identical fins designed according to the simulation model.
Structure	Model rocket engine	Estes C6-5 (ES1507) or equivalent model rocket engine	Select the engine according to the rocket design, weight, and thrust requirements.
Structure	Launch lug	Small guide tube	Inner diameter: 5mm-8mm; ensure it is aligned straight with the rocket body.
Structure	Nose weight	Steel or lead weight	Adjust the weight to shift the centre of gravity (CG) forward for stable flight.